MIL-I-24204A(SHIPS)
15 August 1975
SUPERSEDING
MIL-I-24204(SHIPS)
23 September 1966
(See 6.8)

MILITARY SPECIFICATION

INSULATION, ELECTRICAL, HIGH TEMPERATURE,

BONDED, SYNTHETIC FIBER PAPER

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers flexible electrical insulation used in the coils and slots of electrical machinery at temperatures up to 220° Celsius (C).

1.2 Classification.

1.2.1 Types. The electrical insulation shall be of the following types, as specified (see 6.2):

Type I - Calendered.
Type II - Calendered with improved tear resistance and conformability.

1.2.2 Forms. The electrical insulation shall be of the following forms, as specified (see 6.2):

Form S - Flat sheets, 36 by 36 inches or 24 by 36 inches (see 3.3.2.1). Rolls four inches wide or greater (see 3.3.2.1).
Form T - Tape in rolls less than 4 inches wide (see 3.3.2.2)

1.2.3 Composition. The electrical insulation shall be of high temperature resistant, 100 percent aramid (aromatic polyamide polymer) fiber paper.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
MMM-A-260 - Adhesive, Water-Resistant, (For Sealing Waterproofed Paper).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.
PPP-T-60 - Tape: Packaging, Waterproof.

MILITARY
MIL-P-116 - Preservation-Packaging, Methods of.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

FSC 5970

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 149 - Methods of Test for Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies.
D 150 - Method of Test for A-C Loss Characteristics and Dielectric Constant (Permittivity) of Solid Electrical Insulating Materials.
D 257 - Methods of Test for D-C Resistance or Conductance of Insulating Materials.
D 374 - Methods of Test for Thickness of Solid Electrical Insulation.
D 618 - Methods of Conditioning Plastics and Electrical Insulating Materials for Testing.
D 646 - Methods of Test for Basis Weight of Paper and Paperboard.
D 828 - Method of Test for Tensile Breaking Strength of Paper and Paperboard.
D 1830 - Method of Test for Thermal Stability of Electrical Insulating Coated Fabrics.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INCORPORATED, AGENT
National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P Street, N.W., Washington, D.C. 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. The synthetic fiber paper electrical insulation furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.6).

3.2 Material.

3.2.1 Composition. The synthetic fiber paper shall be made from aramid in the form of short length fibers permanently bound together by fibrous binder particles (fibrids) formed from the same synthetic polymer and containing no extraneous diluents.

3.2.2 Color. The synthetic fiber paper shall be natural in color, normally tan with a mottled appearance which may vary with processing conditions. Variations in color shall not affect the physical, chemical, or electrical properties of the paper.

3.3 Dimensions.

3.3.1 Thickness. The material shall conform to the thickness shown in table I (see 6.2).

3.3.2 Width and length.

3.3.2.1 Form S. Unless otherwise specified (see 6.2), form S material shall be furnished trimmed as either flat sheets or rolls. The dimensions of the flat trimmed sheets shall be

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sizes 36 by 36 inches or 24 by 36 inches, plus or minus 1/4 inch. When form S material is furnished as trimmed rolls, the width tolerances shall be as follows:

Nominal Width (inches)	Tolerance (inches)
4 thru 8	± 1/32
over 8 thru 18	± 1/16
over 18 thru 37	± 1/8
over 37	± 1/4

Table I - Thickness.[1/]

Nominal	Maximum average	Minimum average
Inch	Inch	Inch
0.002	0.0024	0.0016
.003	.0035	.0027
.004	.0046	.0036
.005	.0058	.0045
.007	.0080	.0063
.010	.0110	.0090
.012	.0138	.0102
.015	.0170	.0135
.020	.0220	.0181
.024	.0276	.0204
.030	.0330	.0260

[1/] Type II furnished in 0.007, 0.010, 0.012, and 0.015 thicknesses only.

3.3.2.2 Form T. Unless otherwise specified (see 6.2), tape shall be furnished in rolls in widths of 3/8 to 3-15/16 inches with a tolerance of not greater than plus or minus 1/32 inch. The tape rolls shall have no more than four defects (marked with pressure sensitive polyimide tape having a yellowish appearance) for all thicknesses in standard 11-1/2 inch outside diameter (OD) rolls.

3.4 Performance characteristics. The material shall conform to the quantitative requirements of table II for types I and II material.

3.5 Identification of sheets for stock. Packages of sheets of form S material furnished for stock shall be marked with the nominal thickness and stock number in accordance with MIL-STD-130.

3.6 Directional marking of sheets. The synthetic fiber paper sheets in thicknesses of 0.005 and over shall be marked to show the machine direction. The markings shall be repeated every eight inches across the width of the sheet.

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Table II - Performance characteristics types I and II.

Characteristics	Condition	Unit	Value required[1/] Nominal thickness (inches)									
			0.002	0.003	0.004	0.005	0.007	0.010	0.012	0.015	0.020	0.024	0.030
Apparent density	168/23/55	Grams/cc (minimum average)	0.68	0.74	0.76	0.79	0.82	0.88	0.88	0.90	0.90	0.90	0.90
Dielectric strength	169/23/55	Volts per mil (minimum average)	400	500	500	525	650	650	650	600	600	500	500
	168/23/96	Volts per mil (minimum average)	350	450	450	450	575	575	575	525	550	450	450
	At 220°C Straight	Volts per mil (minimum average)	350	450	450	450	575	575	575	550	550	450	450
	168/23/55 Bent lengthwise	Volts per mil (minimum average)	375	400	400	425	600	600	600	500	475	400	400
	168/23/55 Bent crosswise	Volts per mil (minimum average)	375	400	400	425	600	600	600	500	475	400	400
Volume resistivity	168/23/96	Ohm-cms (minimum average)	1×10^{13}	1×10^{13}	1×10^{13}	1×10^{13}	1×10^{13}	1×10^{13}	1×10^{13}	1×10^{13}	1×10^{13}	1×10^{13}	1×10^{13}
Dissipation factor (60 cycle)	168/23/96	(maximum)	0.03	0.03	0.03	0.03	0.03	0.032	0.032	0.032	0.032	0.035	0.035
	At 220°C	(maximum)	0.03	0.035	0.04	0.04	0.04	0.045	0.050	0.050	0.050	0.058	0.058
Dielectric constant (60 cycle)	168/23/96	(maximum)	3.4	4.0	4.0	4.0	4.2	4.5	4.5	4.5	4.8	5.0	5.0
	At 220°C	(maximum)	3.4	4.0	4.0	4.0	4.2	4.5	4.5	4.5	4.8	5.0	5.0
Tensile strength	168/23/55 Lengthwise	Pounds per inch width (minimum average)	15	28	36	45	72 (60)	110	155 (120)	210 (160)	270	300	330
	168/23/55 Crosswise	Pounds per inch width (minimum average)	8	13	18	28	45 (30)	72 (50)	95 (60)	120 (70)	160	190	240
Tear resistance	168/23/55 Lengthwise	Pounds (minimum average)	1.3	2.0	2.1	2.2	3.0 (5.0)	5.0 (8.0)	6.0 (10)	8.0 (12)	12	13	14
	168/23/55 Crosswise	Pounds (minimum average)	2.0	2.5	3.0	6.0	9.0 (12)	11 (16)	12 (20)	15 (24)	20	24	30
Thermal endurance	As specified in 4.6.9	Temp. °C at 20,000 hours (min)	-	-	-	-	-	220	-	-	-	-	-
Thermal stability	240/300	Volts per mil (minimum average)	325	300	300	300	350	350	350	350	350	350	350

1/ Values for thickness apply to type II only

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the tests specified in table III. Qualification of form S material shall constitute approval for form T material.

4.2.1 Samples for qualification tests. Qualification test samples shall consist of ten square yards of each of the 5, 10, and 20 mil thickness of the type I, form S material and ten square yards of each of the 10 and 15 mil thickness of the type II, form S material.

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Table III - Qualification and quality conformance tests for types I and II.

Test	Requirement paragraph	Test reference	Qualification tests minimum number of determinations per sample unit	Quality conformance inspection minimum number of determinations per sample unit
Visual examination	3.2 and 3.5	4.4.1	As indicated in 4.4.1	As specified in 4.4.1
Dimensions	3.4	4.6.1 and 4.6.2	As specified in 4.6.1 and 4.6.2	As specified in 4.6.1 and 4.6.2
Apparent density	3.4	4.6.3	3	3
Dielectric strength:				
Straight - 168/23/55	3.4	4.6.4	5	-
Straight - 168/23/96	3.4	4.6.4	5	5
Straight at 220°C	3.4	4.6.4	5	-
Bent lengthwise, 168/23/55	3.4	4.6.4	5	-
Bent crosswise, 168/23/55	3.4	4.6.4	5	5
Volume resistivity, 168/23/96	3.4	4.6.5	3	-
Dissipation factor:				
168/23/96	3.4	4.6.6	3	-
At 220°C	3.4	4.6.6	3	3 1/
Dielectric constant:				
168/23/96	3.4	4.6.6	3	-
At 220°C	3.4	4.6.6	3	3 1/
Tensile strength:				
Lengthwise, 168/23/55	3.4	4.6.7	5	5
Crosswise, 168/23/55	3.4	4.6.7	5	5
Tear resistance (slot-cell):				
Lengthwise, 168/23/55	3.4	4.6.8	5	-
Crosswise, 168/23/55	3.4	4.6.8	5	5
Thermal endurance	3.4	4.6.9	As specified in 4.6.9	-
Thermal stability	3.4	4.6.10	-	5 2/

1/ Dissipation factor and dielectric constant shall be measured at 220°C once every year on samples of each type of 0.002 through 0.030 inch material obtained from normal production runs. A copy of the test report shall be kept on file for inspection by authorized Government personnel on demand.

2/ Thermal stability shall be measured once every three years on samples of each type of 0.002 through 0.30 inch material obtained from normal production runs.

4.3 Sampling for quality conformance inspection.

4.3.1 Lot. All material of the same thickness, manufactured under essentially the same conditions and offered for delivery at the same time, shall be considered a lot for purposes of sampling and inspection. The lot size shall be stated in terms of mill rolls. A mill roll is considered a full width machine run roll. Each lot size shall consist of no more than eight mill rolls.

4.3.2 Sampling for visual and dimensional examination. Each roll or package shall be examined as specified in 4.4.1. A roll is considered to be a slit portion of a mill roll and in widths as specified in 3.3.2. A package is considered to be a stack of flat sheets as specified in 3.3.2.

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4.3.3 Sampling for tests. Sample mill rolls from each lot produced shall be selected in accordance with MIL-STD-105, general inspection level II and then subjected to the quality conformance tests (see 4.4.2).

4.4 Quality conformance inspection.

4.4.1 Visual and dimensional examination. Each roll or package shall be examined during cutting operation to verify compliance with this specification. Examination for defects shall be visual or by automatic optical, mechanical, or electrical detector. Dimensional examination shall be as specified in 4.6.2. Any roll or package which contains one or more unmarked visual or dimensional defects shall be considered rejected.

4.4.2 Quality conformance tests. The sample mill roll shall be subjected to the quality conformance tests listed in table III to verify compliance with this specification. Thickness measurements shall be included in quality conformance testing and shall conform to the requirements of 3.3.1. Results of tests shall be averaged and if the average fails to conform with any requirements, this shall be cause for rejection of the mill rolls represented by the sample. The acceptable quality level shall be 6.5 percent defects.

4.5 Conditioning of test specimens.

4.5.1 Nomenclature. The designations indicating conditioning of test specimens shall be in accordance with ASTM D 618.

4.5.2 Temperature and humidity tolerances. Specified temperatures shall be maintained within the tolerances shown in table IV. Specimens shall be tested with moisture contents equal to those obtained in equilibrium with the specified relative humidity plus or minus the tolerances shown in table IV.

Table IV - Temperature and humidity tolerances.

Temperature		Relative humidity	
Celsius (C)	Plus or minus° Celsius (C)	Percent	Plus or minus percent
23 to 180	2	55	5
181 to 300	3		
301 to 325	4	96	3

4.6 Test procedures.

4.6.1 Thickness. Specimen thickness shall be measured using a manual or motorized deadweight dial-type micrometer or its equal in accordance with method C of ASTM D 374. The pressure exerted on the specimen shall be 23 to 27 pounds per square inch (lb/in^2) and the anvil surface upon which the specimen rests shall be 2 inches in diameter for the manual test and 1/4 inch in diameter for the motorized tests. The thickness shall be determined as the average of 10 readings spaced at intervals along the length or across the width of the sample representing the mill roll or rolls being evaluated.

4.6.2 Width and length. Width of rolls, and length and width dimensions of sheets shall be controlled and audited during cutting operations to provide specified sizes within allowable tolerances (see 3.3.2.1). Measurements shall be made on the outside layer of each roll produced, using a steel scale graduated in divisions no larger than one half the allowable tolerance.

4.6.3 Apparent density. Volume and weight of specimens shall be determined after conditioning 168/23/55. Use ASTM D 646 for determining basis weight. Thickness shall be measured in accordance with 4.6.1. The apparent density for each specimen shall be calculated by dividing its weight by its volume.

4.6.4 Dielectric breakdown.

4.6.4.1 Electrical apparatus. Step up transformer, circuit breaker, voltage control and voltmeter shall be in accordance with ASTM D 149. Power supply frequency shall be between 50 and 100 hertz (Hz) and the transformer shall have a rating of not less than 2 kilovoltampere (kVA). The short-time-test at a rate of rise of 500 volts per second shall be used for 7 through 30 mil papers and a rate of rise of 250 volts per second shall be used for 2 through 5 mil papers.

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4.6.4.2 Test electrode assemblies.

4.6.4.2.1 Straight specimens. The test electrodes shall consist of metal cylinder 2 inches in diameter and not less than 1/2 inch in length, with edges rounded to a radius of 1/4 inch. To prevent flash-over, the sample shall extend well away from the outer circumference of the electrodes, or 1/8-inch thick annular rubber or preferably silicone rubber gaskets with 2-1/8 inch diameter center hole shall be employed around the electrodes. The total weight of the upper electrode on the specimen shall be approximately one pound.

4.6.4.2.2 Bent specimens. Brass electrodes, 90 degree V-shaped, as shown in figure 1 shall be used. All edges of the electrodes shall be rounded to approximately 1/64 inch radius.

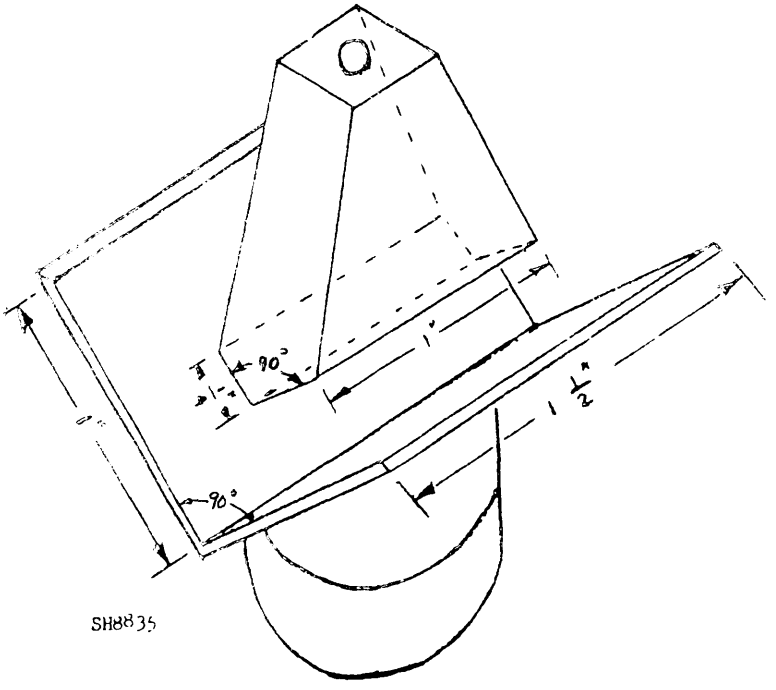


Figure 1 - Dielectric test electrodes 90 degree V-shaped.

4.6.4.3 Bending jig. A 90 degree, V-shaped jig shown in figure 2 shall be used.

4.6.4.4 Procedure.

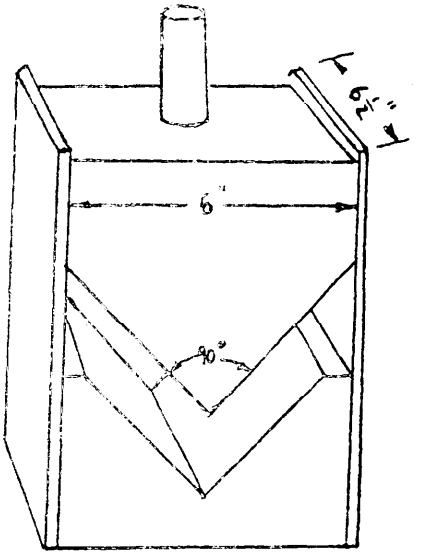
4.6.4.4.1 Straight. Specimen size shall be 5 by 5 inches. Specimens shall be conditioned 168/23/55 and 168/23/96. Dielectric breakdown shall then be measured on the specimens at room conditions of 23°C as rapidly as practicable (within 10 minutes) after removal from the humidity conditioning chamber. The specimens maintained at 220°C for 8 plus or minus 3 minutes shall be subjected to dielectric breakdown at this temperature.

4.6.4.4.2 Bent. Specimens having a minimum size of 3 by 3-1/2 inches shall be cut and conditioned 168/23/55. Specimens shall be bent in the lengthwise direction and others in the crosswise using the bending jig (see 4.6.4.3). Each of the specimens shall then be placed in the V-shaped electrodes (see 4.6.4.2.2) and subjected to dielectric breakdown. A force of 6 pounds shall be exerted by the upper electrode on the specimen.

4.6.4.5 Calculation of dielectric strength. The dielectric strength in volts per mil shall be calculated by dividing the average dielectric breakdown by the average thickness as determined in 4.6.1.

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A the average of the thickness measured at accordance with ASTM D 374, method



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Figure 2 - Bending jig, 90 degree V-shaped.

4.6.5 Volume resistivity.

4.6.5.1 Apparatus. The apparatus shall be in accordance with ASTM D 257 with a direct current potential of 100 to 200 volts applied to the specimen.

4.6.5.2 Test electrodes. Test electrodes shall be conductive rubber electrodes used under pressure with a guarded circuit. The electrodes shall be 1 inch in diameter on both sides. A pressure of 25 lb/in^2 shall be applied to the electrode to insure adequate contact of the electrodes with the specimen.

4.6.5.3 Procedure. The electrodes shall be applied to specimens, 3-1/4 by 3-1/4 inches, and shall be centered with respect to each other. The volume resistance between the electrodes shall be measured at 25°C after voltage has been applied for one minute on each specimen after conditioning 168/23/96.

4.6.5.4 Calculation. Volume resistivity, r, in ohm-centimeters, shall be calculated as follows:

$$r = \frac{RA \times 10^6}{t}$$

where: R = volume resistance in megohms.
A = area of the guarded electrode in square centimeters.
t = average thickness of specimen in centimeters.

4.6.6 Dissipation factor and dielectric constant.

4.6.6.1 Test equipment. Bridge, generator and detector shall be as described in ASTM D 150 for low frequency measurements.

4.6.6.2 Test specimens. Test specimens, 3-1/4 by 3-1/4 inches, shall be used with 1-inch diameter pressure electrodes and a guard circuit as described in 4.6.5.2.

4.6.6.3 Procedure. Specimens shall be conditioned 168/23/96. One specimen at a time shall be removed from the conditioning chamber and measurements shall immediately be made at 23°C, at a frequency of 60 Hz. Other similar specimens will be heated to 220°C for 15 plus or minus 5 minutes and determinations shall be made while at this temperature. Individual dissipation factor and dielectric constant values shall be calculated for each of the specimens, averaged and values reported for each condition. Calculations and edge corrections shall be made in accordance with ASTM D 150.

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4.6.7 Tensile strength.

4.6.7.1 Apparatus. Tensile strength shall be determined on a motor driven pendulum-type or constant rate of jaw separation testing machine.

4.6.7.2 Specimens. The test specimens shall be cut to conform to ASTM D 828. Specimens cut lengthwise and cut crosswise from the sheet shall be measured at 23°C after conditioning 168/23/55.

4.6.7.3 Procedure. Each test specimen shall be placed in the testing machine and stretched until it breaks. The rate of separation of the jaws of the machine shall be not greater than 12 inches plus or minus 1/2 inch per minute. The average tensile strength shall be reported in pounds per inch width.

4.6.8 Tear resistance (slot-cell).

4.6.8.1 Apparatus. The slot-cell tear resistance shall be determined using the tensile strength apparatus specified in 4.6.7.1 with specimens mounted in the slot-cell tear resistance jig shown on figure 3. The tearing edges which come in contact with the specimen have a radius of curvature of 1/32 inch and are set at an angle of 45 degrees. Working drawings of the jig are available from the Naval Ship Research and Development Center, Code 2842, Annapolis, MD 21402.

4.6.8.2 Specimens. Specimens 2-1/2 by 3 inch shall be torn in the tearing jig in the lengthwise direction and in the crosswise direction at 23°C after conditioning 168/23/55. The directions shall be reported as the directions in which the tears run.

4.6.8.3 Procedure. A specimen shall be inserted and alined in the tearing jig. The specimen holder shall be clamped to the upper jaw of the tensile machine. The tearing plate together with the attached guide plate shall then be clamped to the lower jaw. The maximum force, in pounds, developed in tearing through the edges of the specimen extending beyond the holder, at a constant rate of 12 inches per minute, shall be recorded on a chart. The average tear resistance in pounds for each direction of tear shall be calculated and reported.

4.6.9 Thermal endurance. For qualification approval, the thermal endurance evaluation shall consist of tests on the 10 mil thick material only using the method specified in ASTM D 1830, except for the following modifications and additions.

4.6.9.1 Dielectric strength test set. The dielectric strength test set shall be as specified in 4.6.4.1.

4.6.9.2 Test specimens. Test specimens shall be 4-1/2 by 5 inches with the 4-1/2 inch dimension in the lengthwise direction. The specimen shall be inserted in the curved electrode test fixture so that the direction of the bend runs parallel to the 5-inch dimension. Ten sets of 5 specimens shall be used for each aging temperature.

4.6.9.3 Aging specimens. Specimens shall be placed in the oven at room temperature and the temperature then brought up gradually (in about 2 hours) to the desired aging temperature.

4.6.9.4 Exposure temperatures. The specimens shall be exposed at not less than three and preferably four temperatures in the temperature range of 260° to 330°C to establish the life - temperature relationships. The exposure temperatures shall differ by a minimum of 20°C.

4.6.9.5 Thickness. Thickness shall be measured as specified in 4.6.1.

4.6.9.6 End point calculation. The end point to determine life shall be 300 volts per mil. The straight line representing the life of the material as a function of temperature shall be extrapolated to 20,000 hours. The temperature at which the life line crosses the 20,000 hours ordinate shall also be recorded.

4.6.10 Thermal stability.

4.6.10.1 Test specimens. Five specimens shall be prepared as described in 4.6.9.2 for each thickness of material.

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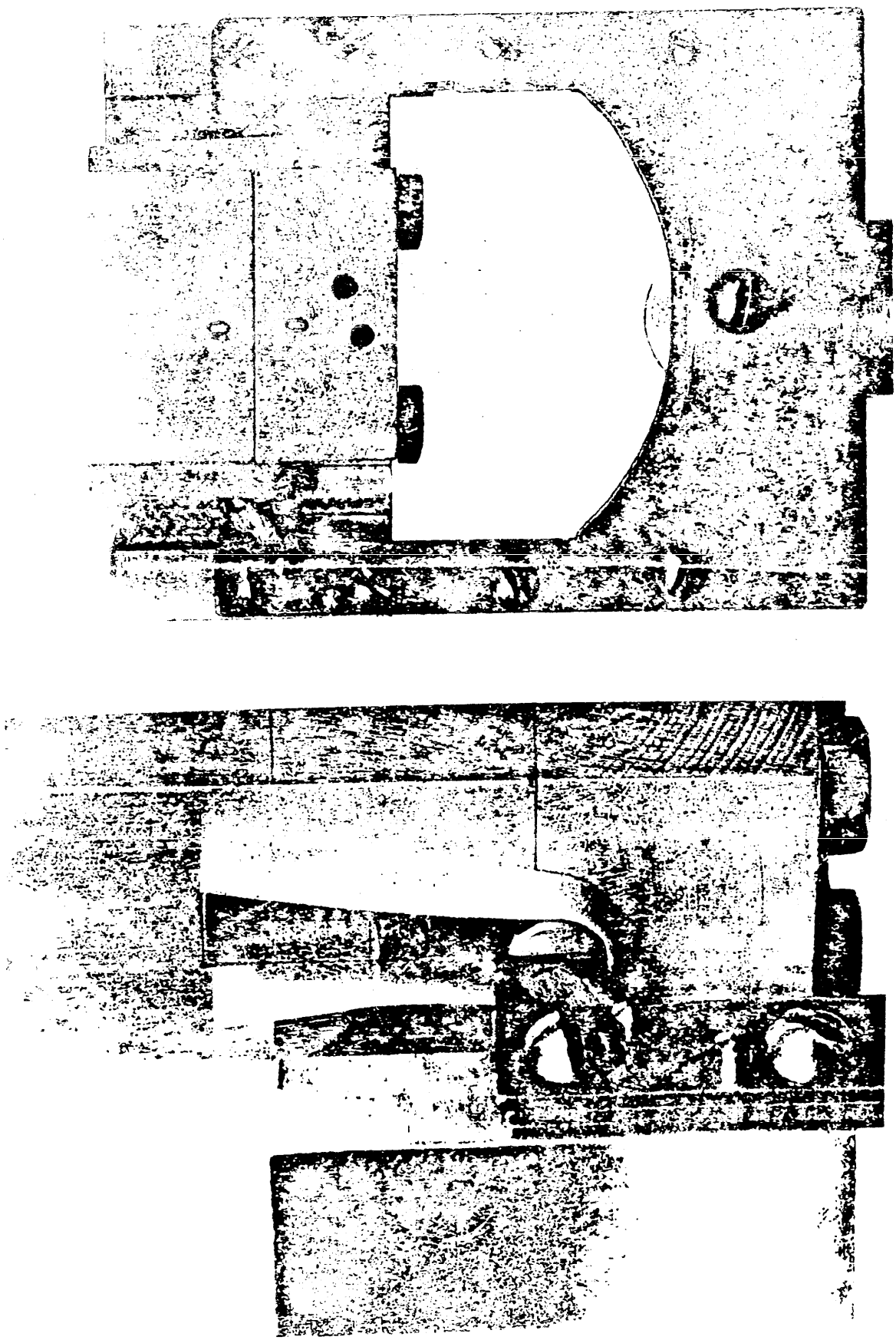


Figure 3 - Slot-cell tear resistance jig specimen in position to start tear (two views).

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4.6.10.2 Procedure. The specimens will be aged for 10 days at 300°C. Specimens shall then be removed from the oven, cooled and tested for dielectric breakdown in accordance with 4.6.9. The dielectric strength in volts per mil shall be calculated by dividing the average dielectric breakdown by average thickness as determined in 4.6.1.

4.7 Inspection of preparation for delivery. The packaging, packing and marking shall be examined to determine compliance with requirements of Section 5 of this specification.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Form S insulation sheets. Form S insulation sheets shall be packaged in accordance with method IC of MIL-P-116 as follow: The sheets shall be stacked in like thickness and size in quantities specified (see 6.2) with a sheet of fiberboard the same size as the sheets, placed above and under the stack, secured in place with tape conforming to type PPP-T-60. The stacks shall be individually wrapped with waterproof barrier material conforming to type PPP-B-1055 with all folds, seams, and laps sealed with water resistant adhesive conforming to MMM-A-260.

5.1.1.2 Form T insulation tape. Form T insulation tape shall be individually packaged in accordance with method IC-3 of MIL-P-116. Packaged rolls of tape shall be individually packaged in folding or set-up boxes conforming to PPP-B-566 or PPP-B-676. Each box shall be closed and sealed in accordance with the box specification or appendix thereto.

5.1.1.3 Intermediate package. Form T insulation tape of like description and size packaged as specified in 5.2.1.1.1.2 shall be intermediate packaged in snug-fitting fiberboard boxes conforming to PPP-B-636 class weather resistant. Boxes shall be closed in accordance with method V of the appendix to the box specification. The gross weight shall not exceed the weight limitation of the box specification.

5.1.2 Level C. Packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. The suppliers normal retail or wholesale packaging methods maybe utilized when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The insulation, packaged as specified (see 6.2), shall be packed in overseas type, wood cleated fiberboard, nailed wood, fiber, wirebound wood, wood cleated veneer paper overlaid, or wood cleated plywood boxes conforming to PPP-B-591, PPP-B-621, PPP-B-636, class weather resistant, PPP-B-585, PPP-B-576, class 2, or PPP-B-601, respectively, at the option of the contractor. Shipping containers, except PPP-B-636 boxes, shall have case liners conforming to MIL-L-10547. Case liners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto, except PPP-B-636 fiber boxes shall be closed method V in accordance with the appendix to the box specification. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification. Unit or intermediate fiber boxes conforming to PPP-B-636 class weather-resistant-closed as specified herein and used as the shipping container, need not be overpackaged.

5.2.2 Level B. The insulation, packaged as specified (see 6.2), shall be packed in domestic type wood cleated fiberboard, nailed wood, wirebound wood, cleated plywood, wood cleated veneer paper overlaid boxes, or fiberboard boxes conforming to PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-601, PPP-B-576, or PPP-B-636, respectively at the option of the supplier. Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight, of wood or wood cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification. Intermediate fiber boxes conforming to PPP-B-636, closed as specified herein, and used as the shipping container need not be overpacked.

5.2.3 Level C. The insulation, packaged as specified (see 6.2), shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with Uniform Freight or National Motor Freight classification rules or regulations or other carrier rules as applicable to the mode of transportation.

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5.3 Marking. In addition to any special marking required herein or by the contract or order (see 6.2), unit and intermediate packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6 NOTES

6.1 Intended use. The material covered by this specification is intended for use in generator, motor and transformer coils, slot liners and ground insulation, phase to phase insulation, phase and end insulation, primary-secondary separation, coil exteriors, coil wrapper tapes, and similar applications. The highest temperature (limiting "hottest spot" temperature) to which the insulation may be subjected to continuously with normal life expectancy is 220°C. The material should not be used in applications where there is a known corona problem. In general, operating voltage stress should be below 50 volts per mil of thickness. Type I material is for general purpose use; type II material is for use where more tear resistance is required and where increased conformability or pull down on irregular surfaces is required.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Form required (see 1.2).
(d) Nominal thickness (see table I).
(e) Whether or not form S material is to be furnished trimmed as either flat sheets or rolls (see 3.3.2.1).
(f) Width and length of form S material, (see 3.3.2.1).
(g) Width and length of rolls of form T material (see 3.3.2.2 and 6.3).
(h) Quantity square yards (see 6.4).
(i) Levels of packaging and packing required (see 5.1 and 5.2).
(j) If special marking is required (see 5.3).

6.3 Rolls of standard width and put-up.

(a) Widths: 12, 24, 36, and 72 inches.
(b) Standard put-up is on 3" ID core with 11-1/2 ± 1/2" OD for both form S and form T rolls.
(c) The approximate linear yards contained on rolls of standard put-up is shown in table V.

Table V - Lengths on standard rolls.

Nominal insulation thickness	Approx. linear yards 3" Core and 11-1/2" OD roll
Inch	
0.002	1250
.003	840
.004	640
.005	480
.007	350
.010	260
.012	200
.015	170
.020	130
.024	110
.030	90

6.4 Sheets of standard sizes and put-up, Form S

(a) Sizes: 36 x 36 inches and 24 x 36 inches.
(b) Sheets per carton is shown in table VI.

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Table VI - Sheets per carton.

Nominal thickness	Sheets per carton	
	36 x 36	24 x 36
.002	1400	2000
.003	900	1300
.004	700	900
.005	500	550
.007	300	500
.010	200	300
.012	200	300
.015	150	200
.020	100	150
.024	100	100
.030	50	100

6.5 Approximate areas per pound. The approximate areas per pound of the synthetic fiber paper covered by this specification are shown in table VII.

Table VII - Areas per pound (approximate).

Nominal insulation thickness	Area in square yards per pound
Inch	
0.002	13.7
.003	8.7
.004	6.5
.005	4.9
.007	3.2
.010	2.2
.012	1.8
.015	1.4
.020	1.0
.024	0.8
.030	0.7

6.6 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL-24204 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.7).

6.7 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.8 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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OMB Approval
No. 22-R255

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MATERIAL PROCURED UNDER A

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A. GIVE PARAGRAPH NUMBER AND WORDING

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